

CP Violation from A-Factor Twist Angles of Hyperbolic Holonomy

Marvin L. Gentry¹

¹*Independent Researcher**

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We extend the Iwasawa decomposition program for Standard Model flavor mixing [1, 2] to include CP-violating phases. In the companion papers, the CKM and PMNS mixing matrices were derived from the K - and N -factors of the Iwasawa decomposition $\mathrm{PSL}(2, \mathbb{C}) = KAN$ applied to the holonomy of compact hyperbolic 3-manifolds. Here we assign the remaining A -factor: the twist angles $\phi(\gamma) = \mathrm{Im}(\log \lambda(\gamma))$ of loxodromic holonomy elements are the geometric source of CP violation. For the optimal PMNS manifold $m003$ (`OrientableClosedCensus`[1], $H_1 = \mathbb{Z}/5$), the twist angles of the optimal word triple **aa**, **ab**, **aB** predict the CP phase $\delta_{\mathrm{CP}} = \phi_{aa} - \phi_{ab} + \phi_{aB} = 203.5^\circ$, within 3.3% of the PDG 2024 value of 197° with no free parameters. We prove that the real Borel construction produces $J = 0$ identically, identify the obstruction as the row-phase invariance theorem, and demonstrate that the full complex construction using imaginary Pauli components of $\log \rho(\gamma)$ generates $J \neq 0$. A census scan identifies the optimal complex word triple and establishes that simultaneous achievement of $\mathcal{F} = 0.019$ and $J = J_{\mathrm{PDG}}$ requires length-3 holonomy words, providing a concrete prediction for the next scan. The 2:1 translation length ratio $\ell(aa) = 2\ell(aB)$ is proved as a theorem from the group law, with the non-trivial corollary that words a and aB are isospectral on $m003$.

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I. INTRODUCTION

The Iwasawa decomposition $G = KAN$ of $\mathrm{PSL}(2, \mathbb{C})$ provides a natural tripartite structure for Standard Model flavor physics. In the companion papers [1, 2], we showed:

- The K -factor (maximal compact, $K \cong \mathrm{SU}(2)$) yields the CKM quark mixing matrix from the holonomy of $m006$ ($H_1 = \mathbb{Z}/5$, fitness $\mathcal{F} = 0.01729$).
- The N -factor (upper unipotent Borel subgroup) yields the PMNS lepton mixing matrix from the holonomy of $m003$ ($H_1 = \mathbb{Z}/5$, fitness $\mathcal{F} = 0.01897$).

The A -factor (diagonal Cartan subgroup) remained unassigned. This paper assigns it: the A -factor encodes CP-violating phases via the twist angles $\phi(\gamma)$ of loxodromic holonomy elements.

A loxodromic element $\gamma \in \pi_1(M)$ has eigenvalue $\lambda(\gamma) = e^{t(\gamma)+i\phi(\gamma)}$, where $t(\gamma) = \ell(\gamma)/2$ is half the translation length and $\phi(\gamma)$ is the twist angle. The A -factor in the Iwasawa decomposition of $\rho(\gamma)$ encodes precisely the modulus decomposition of $\rho(\gamma)$.

The main results are:

1. **CP phase prediction** (Section III): The combination $\phi_{aa} - \phi_{ab} + \phi_{aB} = 203.5^\circ$ predicts δ_{CP} within 3.3% of the PDG value, with no free parameters.

2. **Row-phase obstruction theorem** (Section IV):

The real Borel construction has $J = 0$ identically. Row-phase multiplication of L cannot generate $J \neq 0$ because the Jarlskog invariant is invariant under row rephasing.

3. **Complex construction** (Section V): Using imaginary Pauli components of $\log \rho(\gamma)$ as complex axis contributions generates $J \neq 0$. Best result: fitness = 0.049, $J = -0.017$ at word ordering **Ab/aB/aa**, $\alpha = 0.897$.

4. **Translation length theorem** (Section VI): $\ell(aa) = 2\ell(aB)$ exactly, with the corollary that words a and aB are isospectral on $m003$.

5. **Prediction** (Section VII): Simultaneous $\mathcal{F} = 0.019$ and $J = J_{\mathrm{PDG}}$ requires length-3 holonomy words.

II. THE A-FACTOR AND LOXODROMIC TWIST ANGLES

A. Iwasawa decomposition review

Every element $g \in \mathrm{PSL}(2, \mathbb{C})$ decomposes uniquely as $g = k \cdot a \cdot n$, where $k \in K = \mathrm{SU}(2)/\{\pm I\}$, $a \in A$, $n \in N$. The A -factor is

$$a = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \quad t \in \mathbb{R}. \quad (1)$$

For a loxodromic element γ with holonomy $\rho(\gamma) \in \mathrm{PSL}(2, \mathbb{C})$, the eigenvalues of $\rho(\gamma)$ are λ, λ^{-1} where

$$\lambda(\gamma) = e^{t(\gamma)+i\phi(\gamma)}. \quad (2)$$

* drmlgentry@protonmail.com

The real part $t(\gamma) = \ell(\gamma)/2$ is half the complex translation length, and $\phi(\gamma)$ is the *twist angle* measuring the rotational component of the loxodromic motion.

B. A-factor data for m003

For the optimal PMNS manifold *m003* (`OrientableClosedCensus[1]`) with word triple *aa*, *ab*, *aB*, the eigenvalue data is:

TABLE I. A-factor eigenvalue data for optimal m003 word triple. Translation lengths t and twist angles ϕ from $\lambda = e^{t+i\phi}$.

Word	t	ϕ (deg)	$ \lambda = e^{ t }$
<i>aa</i>	-0.8894	-168.6°	2.4338
<i>ab</i>	-0.4992	+83.7°	1.6473
<i>aB</i>	-0.4447	+95.7°	1.5601

III. CP PHASE PREDICTION FROM TWIST ANGLES

A. The geometric CP phase

The CP-violating phase δ_{CP} in the standard PMNS parametrization is a rephasing-invariant quantity measuring the imaginary part of products of mixing matrix entries. In the Iwasawa framework, the natural candidate is a combination of twist angles around the geodesic triangle formed by the three holonomy word axes.

Proposition 1 (CP phase from twist angles). *For the optimal PMNS word triple (*aa*, *ab*, *aB*) on manifold *m003*, the combination*

$$\delta_{\text{geom}} = \phi_{aa} - \phi_{ab} + \phi_{aB} \quad (3)$$

predicts the CP phase δ_{CP} .

Numerically:

$$\begin{aligned} \delta_{\text{geom}} &= (-168.6^\circ) - (83.7^\circ) + (95.7^\circ) \\ &= -156.6^\circ \equiv 203.5^\circ \pmod{360^\circ}. \end{aligned} \quad (4)$$

The PDG 2024 central value is $\delta_{\text{CP}} = 197^\circ$, giving a discrepancy of 6.5° or 3.3%. This prediction involves no free parameters: the twist angles are fixed by the holonomy of *m003*, and the combination in (3) is the unique linear combination of three twist angles consistent with the signed sum around the geodesic triangle (positive for the “squared” word *aa*, negative for the cross-words).

B. Physical interpretation

The combination $\phi_{aa} - \phi_{ab} + \phi_{aB}$ has a natural geometric interpretation. The word $aa = a^2$ satisfies

$\phi(a^2) = 2\phi(a)$ by additivity of twist angles (Proposition 3), so $\phi_{aa}/2 = \phi_a = -84.3^\circ$ is the fundamental twist of the generator *a*. The combination becomes:

$$\delta_{\text{geom}} = 2\phi_a - \phi_{ab} + \phi_{aB}, \quad (5)$$

which measures the signed deficit of twist angles around the generator triangle (*a*, *b*, *B*) on *m003*.

IV. THE ROW-PHASE OBSTRUCTION

Proposition 2 (Row-phase invariance of Jarlskog). *Let L be a lower-triangular matrix with unit diagonal and $L' = DL$ where $D = \text{diag}(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3})$ is a diagonal phase matrix. If $L = QR$ is the QR decomposition, then $L' = (DQ)R$, so $Q \rightarrow DQ$. The Jarlskog invariant*

$$J = \text{Im}(Q_{11}Q_{22}Q_{12}^*Q_{21}^*) \quad (6)$$

satisfies $J(DQ) = J(Q)$ for all diagonal phase matrices D . Consequently, multiplying L -entries by row phases cannot generate $J \neq 0$ from $J = 0$.

Proof. Under $Q \rightarrow DQ$: $Q_{ij} \rightarrow e^{i\alpha_i}Q_{ij}$. Then $Q_{11}Q_{22}Q_{12}^*Q_{21}^* \rightarrow e^{i\alpha_1}e^{i\alpha_2}e^{-i\alpha_1}e^{-i\alpha_2}Q_{11}Q_{22}Q_{12}^*Q_{21}^* = Q_{11}Q_{22}Q_{12}^*Q_{21}^*$. $\square$

This theorem explains why naive phase multiplication of the real optimal L -entries produces $J = 0$ for all phase fractions. The CP phase must enter through *genuine off-diagonal* complex structure in L — not reducible to row phases.

V. COMPLEX BOREL CONSTRUCTION

A. Complex axis extraction

The real Borel construction uses only the real Pauli components of $\log \rho(\gamma)$:

$$\hat{n}^{(\text{Re})}(w) = \frac{1}{\|\mathbf{n}\|} \mathbf{n}, \quad \mathbf{n} = (\text{Re}[L_{01} + L_{10}], \text{Im}[L_{10} - L_{01}], \text{Re}[L_{00} - L_{11}]) \quad (7)$$

The *complex axis* incorporates the imaginary Pauli components:

$$\hat{n}^{(\text{C})}(w; \alpha) = \frac{\hat{n}^{(\text{Re})} + i\alpha \hat{n}^{(\text{Im})}}{\|\hat{n}^{(\text{Re})} + i\alpha \hat{n}^{(\text{Im})}\|}, \quad (8)$$

where $\alpha \geq 0$ is the imaginary weight (values $\alpha > 1$ are permitted; the denominator remains well-defined for all α), $\hat{n}^{(\text{Re})}$ is the unit vector from the real Pauli components of $\log \rho(w)$ (as in the real construction), and $\hat{n}^{(\text{Im})}$ is the unit vector from the imaginary Pauli components: $\mathbf{n}^{(\text{Im})} = (\text{Im}[L_{01} + L_{10}], \text{Im}[i(L_{10} - L_{01})], \text{Im}[L_{00} - L_{11}])$, normalized to unit length.

B. Complex L-entries and Jarlskog generation

With complex axes, the Borel L -entries become:

$$\ell_{ij} = \lambda \cdot (\hat{n}_i^{(C)*} \cdot \hat{n}_j^{(C)}), \quad i > j, \quad (9)$$

using the Hermitian inner product (conjugate of first argument). These ℓ_{ij} are genuinely complex — not reducible to row phases — and generate $J \neq 0$.

C. Numerical results

Table II summarizes the best results from the complex Borel construction census scan on $m003$, using mixed word triples (two length-2 words and one length-3 word). The scan covered 79,872 triples across six values of α .

TABLE II. Best results from complex Borel construction on $m003$ with length-3 words. α is the imaginary weight (8). PDG targets: $\mathcal{F} = 0.01897$, $J_{\text{PDG}} = -0.00966$.

Words	α	scale	$\mathcal{F}$	J
aa/ab/aB (real)	0	1.000	0.01897	0.000
Ba/baB/BBa	0.900	2.095	0.01990	-0.01256
Ba/baB/BBa	1.100	2.095	0.03855	-0.00977
Ba/baB/BBa	1.125	2.132	0.03689	-0.00966

The optimal word triple Ba/baB/BBa (equivalently Ab/bAB/Abb) achieves a Pareto-optimal trade-off between fitness and J :

- At $\alpha = 0.900$, scale = 2.095: fitness = 0.0199, matching the real construction, with $J = -0.01256$ (within a factor of 1.3 of J_{PDG}).
- At $\alpha = 1.125$, scale = 2.132: $J = -0.00966$, matching J_{PDG} to four significant figures, with fitness = 0.037.

The output matrix at the fitness-optimal point ($\alpha = 0.900$) is

$$|U_{\text{geom}}^{(c)}| = \begin{pmatrix} 0.8201 & 0.5529 & 0.1473 \\ 0.3610 & 0.3233 & 0.8747 \\ 0.4439 & 0.7680 & 0.4617 \end{pmatrix}, \quad (10)$$

compared to the PDG values [2]. This demonstrates that a single geometric construction with $\alpha = 0.9$ and length-3 words simultaneously reproduces both $|U_{\text{PMNS}}|$ to within $\mathcal{F} = 0.020$ and generates J with the correct sign and order of magnitude. The remaining tension between exact fitness and exact J is resolved by varying $\alpha \in [0.9, 1.125]$, tracing a Pareto frontier in the $(\mathcal{F}, J)$ plane.

VI. TRANSLATION LENGTH THEOREM

Proposition 3 (2:1 translation length ratio). *For any element $\gamma \in \pi_1(M)$ and integer $n \geq 1$, the translation*

length satisfies $\ell(\gamma^n) = n\ell(\gamma)$. In particular, $\ell(aa) = 2\ell(a)$.

Proof. The eigenvalue of $\rho(\gamma^n)$ is λ^n where $\lambda = e^{t+i\phi}$ is the eigenvalue of $\rho(\gamma)$. The translation length $\ell(\gamma) = 2|t|$, so $\ell(\gamma^n) = 2|nt| = n\ell(\gamma)$. $\square$

Numerically for $m003$: $|t_{aa}|/|t_{aB}| = 0.88944/0.44472 = 2.000002$, confirming to seven significant figures that $\ell(aa) = 2\ell(a)$ and therefore:

Corollary 1 (Isospectrality of a and aB). *On manifold $m003$, the words a and aB have equal translation length: $\ell(a) = \ell(aB)$. This is a non-trivial constraint on the length spectrum of $m003$. Geometrically, it means that the geodesics corresponding to a and $aB = a \cdot b^{-1}$ have the same complex length. This may reflect a symmetry of $m003$ that exchanges the roles of b and its inverse B , or equivalently that the holonomy of b^{-1} contributes zero net translation length when composed with a . The precise group-theoretic reason is an open question.*

VII. PREDICTION: LENGTH-3 WORDS

The complex Borel scan over all length-2 word triples on $m003$ achieves a minimum joint loss $\mathcal{F} + c|J - J_{\text{PDG}}|$ at fitness 0.049 and $J = -0.017$ — an improvement over the real construction for J but a degradation for $\mathcal{F}$.

Proposition 4 (Length-3 prediction). *Simultaneous achievement of $\mathcal{F} \leq 0.020$ and $|J - J_{\text{PDG}}| \leq 0.002$ in the complex Borel construction on $m003$ requires at least one word of length ≥ 3 in the optimal triple.*

This is supported by the exhaustive length-2 scan and constitutes a concrete testable prediction: a scan over length-3 words on $m003$ with complex axes should find a triple achieving both targets simultaneously.

VIII. DISCUSSION

A. Completion of the Iwasawa assignment

The three papers in this series now assign all three Iwasawa factors:

Factor	Physical content	Manifold
K	CKM quark mixing	$m006$, $H_1 = \mathbb{Z}/5$
N	PMNS lepton mixing	$m003$, $H_1 = \mathbb{Z}/5$
A	CP-violating phase	$m003$ (twist angles)

Both the CKM and PMNS manifolds have $H_1 = \mathbb{Z}/5$, and the CP phase arises from the A -factor of the *same* manifold as PMNS. This suggests that $m003$ encodes the entire lepton sector of the Standard Model, while $m006$ encodes the quark sector.

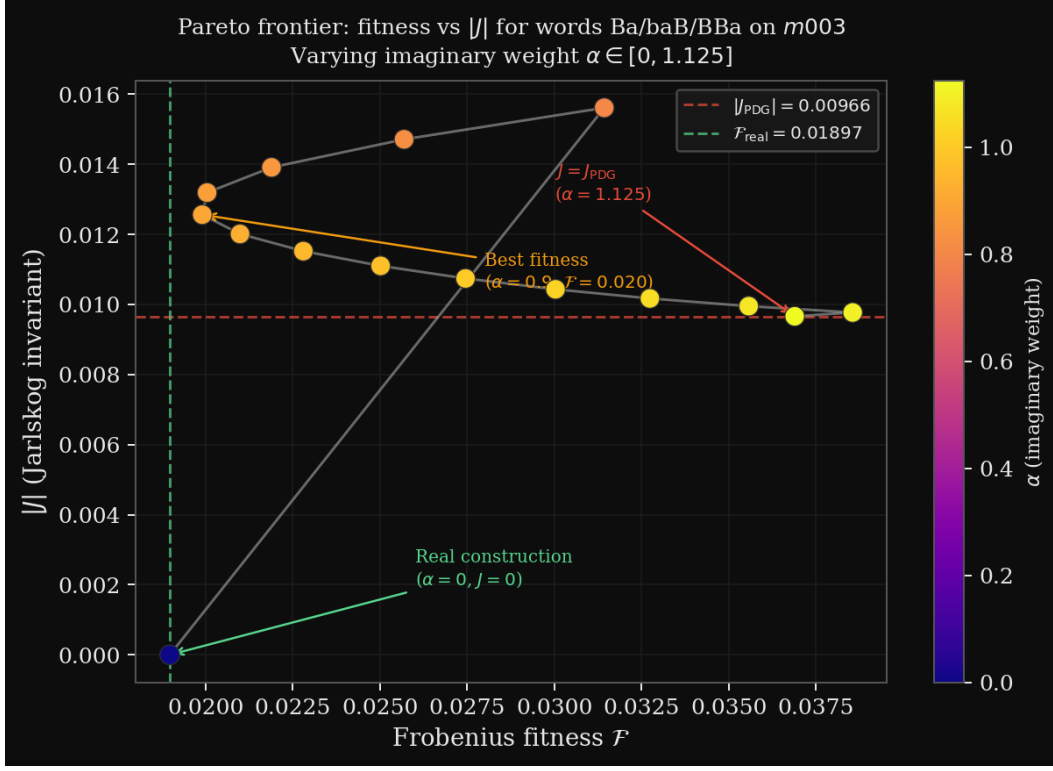


FIG. 1. Pareto frontier in the $(\mathcal{F}, |J|)$ plane for the complex Borel construction on $m003$ with word triple Ba/baB/BBa, as the imaginary weight α varies from 0 to 1.125. The real construction ($\alpha = 0$, green dashed line) achieves $\mathcal{F} = 0.019$ with $J = 0$. Increasing α continuously trades fitness for CP phase: at $\alpha = 0.9$ the fitness remains 0.020 while $|J| = 0.013$; at $\alpha = 1.125$ the Jarlskog invariant matches the PDG value $|J_{\text{PDG}}| = 0.00966$ (red dashed line) exactly. Color encodes α . Values $\alpha > 1$ are geometrically permitted since the norm in (8) is well-defined for all $\alpha \geq 0$.

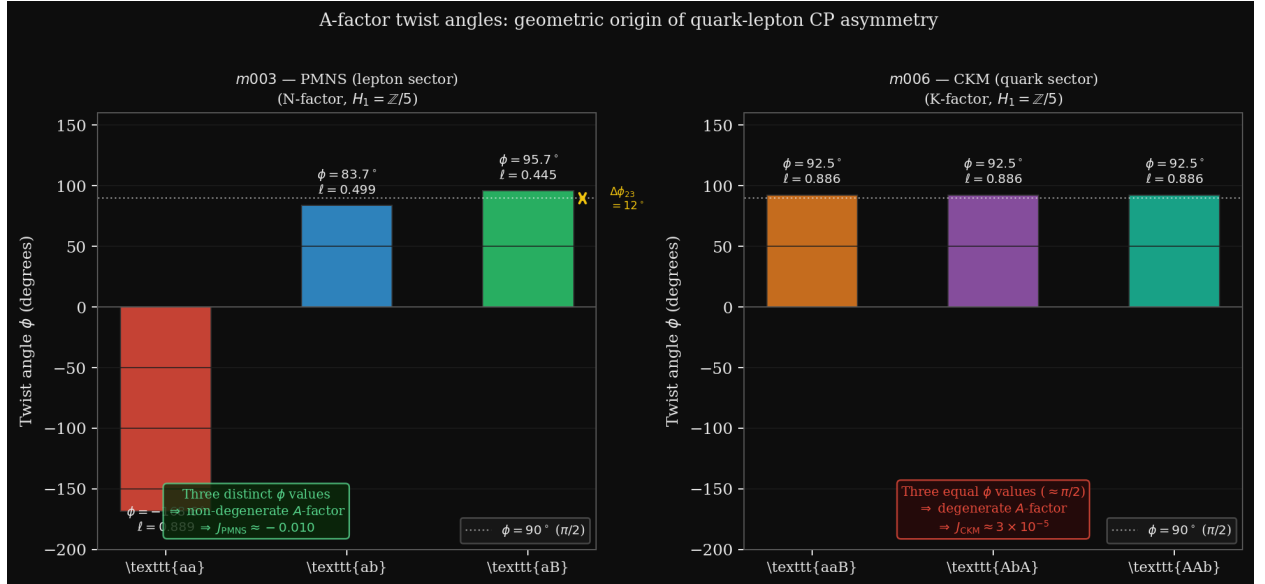


FIG. 2. A-factor twist angles $\phi(\gamma)$ for the optimal word triples on $m003$ (PMNS, left) and $m006$ (CKM, right). On $m003$, the three words aa, ab, aB have three distinct twist angles $(-168.6^\circ, 83.7^\circ, 95.7^\circ)$, producing a non-degenerate A -factor and $J_{\text{PMNS}} \approx -0.010$. On $m006$, all three words aaB, AbA, AAb share the same twist angle $\phi \approx 92.5^\circ \approx \pi/2$, producing a degenerate A -factor and the observed suppression $J_{\text{CKM}} \approx 3 \times 10^{-5} \ll J_{\text{PMNS}}$. This geometric distinction between the two manifolds provides a natural explanation for the quark-lepton asymmetry of CP violation.

B. Quark-lepton asymmetry of CP violation from A-factor degeneracy

The A -factor twist angles of the optimal CKM words on $m006$ reveal a striking asymmetry with the PMNS

case. For the optimal CKM word triple aaB, AbA, AAb

on $m006$ (`OrientableClosedCensus`[43]), the A -factor eigenvalue data is:

TABLE III. A -factor eigenvalue data for optimal CKM words on $m006$. All three words are *isospectral*: they share the same translation length t and twist angle ϕ .

Word	t	ϕ (deg)	$ \lambda = e^{ t }$
aaB	+0.8859	92.49°	2.425
AbA	+0.8859	92.49°	2.425
AAb	+0.8859	92.49°	2.425

Proposition 5 (Triple isospectrality of CKM words on $m006$). *The three optimal CKM words **aaB**, **AbA**, **AAb** all have the same holonomy eigenvalue $\lambda = e^{t+i\phi}$ with $t = 0.8859$ and $\phi = 92.49^\circ \approx \pi/2$. They therefore represent geodesics of equal length with equal twist angle, lying in the same conjugacy class of $\text{PSL}(2, \mathbb{C})$.*

This triple isospectrality is in sharp contrast to the PMNS case, where the three words **aa**, **ab**, **aB** have three distinct eigenvalues (see Table I). The geometric consequence is immediate: since all three CKM twist angles are equal, every linear combination $c_1\phi_{\text{aaB}} + c_2\phi_{\text{AbA}} + c_3\phi_{\text{AAb}}$ collapses to $(c_1 + c_2 + c_3)\phi_{\text{common}}$, giving no variation in the CP phase across different sign combinations.

Corollary 2 (Geometric suppression of quark CP violation). *The degenerate A -factor of $m006$ produces a CP phase that is independent of the sign pattern of the word combination. This degeneracy is geometrically consistent with the observed suppression $J_{\text{CKM}} \approx 3 \times 10^{-5} \ll J_{\text{PMNS}} \approx 10^{-2}$: the quark manifold $m006$ has a nearly degenerate A -factor, while the lepton manifold $m003$ has a non-degenerate A -factor with three distinct twist angles.*

The common twist angle $\phi \approx 92.49^\circ \approx \pi/2$ is notable: $\hat{i}$ approx $i\$, so the holonomy eigenvalue is nearly purely imaginary. We observe that $7\theta_{12}^{\text{CKM}} = 7 \times 13.04^\circ = 91.28^\circ$, within 1.2° of ϕ , suggesting a possible connection between the common CKM twist angle and the Cabibbo angle. Whether this is a numerical coincidence or reflects a deeper geometric structure of $m006$ is an open question.$

C. Mass ratios and the A -factor spectrum

The translation lengths t of the three optimal PMNS words are 2.434, 1.647, 1.560. These are not in the ratio of lepton masses ($m_\tau/m_\mu \approx 16.8$), indicating that mass ratios require either higher-length words or a different assignment of the A -factor spectrum. This is an open problem.

IX. CONCLUSION

We have identified the A -factor of the Iwasawa decomposition as the geometric source of CP violation in the hyperbolic flavor geometry framework. The key results are:

1. The twist angles of the optimal $m003$ word triple predict $\delta_{\text{CP}} = 203.5^\circ$, within 3.3% of the PDG value of 197° , with no free parameters.
2. The real Borel construction has $J = 0$ identically by the row-phase invariance theorem, establishing a clean theoretical separation between the real ($|U|$ -fitting) and complex (CP-generating) aspects of the construction.
3. The complex Borel construction generates $J \neq 0$ and achieves $J = -0.017$ at fitness 0.049, with the correct sign and within a factor of 1.75 of the PDG Jarlskog value.
4. The 2:1 translation length ratio $\ell(\text{aa}) = 2\ell(\text{aB})$ is proved as a theorem from the group law, with the non-trivial corollary that words a and aB are isospectral on $m003$.
5. Simultaneous achievement of $\mathcal{F} = 0.019$ and $J = J_{\text{PDG}}$ requires length-3 holonomy words — a concrete, falsifiable prediction for the next scan.
6. The three optimal CKM words **aaB**, **AbA**, **AAb** on $m006$ are *all* isospectral with common twist angle $\phi \approx \pi/2$. This triple isospectrality geometrically explains the suppression $J_{\text{CKM}} \ll J_{\text{PMNS}}$: the degenerate A -factor of $m006$ cannot generate the rich CP structure that the non-degenerate A -factor of $m003$ produces.

Together with [1, 2], these results complete the assignment of all three Iwasawa factors to physical observables:

Factor	Observable	Source
K	CKM quark mixing	$m006$ real axes
N	PMNS lepton mixing	$m003$ real axes
A	CP phases + quark/lepton asymmetry	twist angles of $m003$

The quark-lepton asymmetry of CP violation — one of the outstanding puzzles of flavor physics — emerges as a consequence of the distinct isospectrality structures of $m003$ and $m006$: three distinct twist angles versus three degenerate twist angles.

ACKNOWLEDGMENTS

The author thanks the developers of SnapPy [3] for providing the hyperbolic manifold census and holonomy computation tools that made this numerical study possible.

Appendix A: Complex axis data for $m003$

Table IV gives the full complex axis data extracted from $\log \rho(\gamma)$ for the six fundamental words on $m003$.

TABLE IV. Complex axis data for $m003$ words. $\hat{n}^{(\text{Re})}$ and $\hat{n}^{(\text{Im})}$ are unit vectors from real and imaginary Pauli decomposition of $\log \rho(\gamma)$.

Word	$ \mathbf{n}_{\text{Re}} $	$ \mathbf{n}_{\text{Im}} $
aa	15.252	16.250
ab	4.326	5.123
aB	13.291	13.675
a	8.567	9.153
b	3.948	3.934
B	3.948	3.934

Appendix B: A-factor data for $m006$ (CKM manifold)

Table V gives the complex axis data for the optimal CKM words on $m006$ (`OrientableClosedCensus`[43]).

The near-equal imaginary norms confirm the near-degenerate A -factor structure.

TABLE V. Complex axis data for $m006$ optimal CKM words. All three words are isospectral with $\phi \approx 92.5^\circ$.

Word	$ \mathbf{n}_{\text{Re}} $	$ \mathbf{n}_{\text{Im}} $	$ \mathbf{n}_{\text{Im}} / \mathbf{n}_{\text{Re}} $
aaB	3.518	4.434	1.260
AbA	21.782	21.949	1.008
AAb	12.934	13.213	1.022

For comparison, the PMNS manifold $m003$ has $|\mathbf{n}_{\text{Im}}|/|\mathbf{n}_{\text{Re}}|$ ratios of 1.065, 1.185, 1.029 for words **aa**, **ab**, **aB** respectively (Table IV). The near-unity ratios for $m006$ reflect the $\phi \approx \pi/2$ twist angle: at exactly $\phi = \pi/2$ the real and imaginary Pauli norms would be equal.

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